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linkedin.com/in/adityavardhanmishra • github.com/aaaaaaaaaaaaavm/EMOCD
BTech Mechanical Engineering, Symbiosis Institute of Technology (SIT), Pune • 2023–2027

Working on electromagnetic satellite deployment. Five years on one question — why CubeSats are still deployed with springs — now a complete design study with a published defect record, an independently cross-checked field model, and every claim traceable to the script that produced it. Also hands-on: engine rebuilds, ECU calibration, 2nd globally at IREC 2025. Seeking propulsion, structures, or test & validation work. **Recognised by the ISRO Chairman** for STEM outreach • Space Generation Advisory Council.

Space systems

Electromagnetic Orbital CubeSat Deployer (EMOCD)

Independent, final-year project • IEEE manuscript in preparation 2021–present

- Architected a magazine-fed **ironless double-sided Halbach linear synchronous motor** that ejects **unmodified** 3U CubeSats at **16.5 m/s** (13.1–18.5 m/s across the 1U–12U family) at **10.7 g** on **2.80 kJ** per shot — eight times a spring, inside standard CubeSat qualification loads.
- Selected the LSM over a coilgun in a documented architecture trade: a coilgun needs a conductive armature on the customer satellite, and its velocity advantage is capped by the payload's own g-limit. Verified against published work reaching 321 m/s at ∼1350 g — roughly 100× CubeSat qualification.
- Derived the thrust constant **11.22 N per kA/m** from a winding-resolved model, then **wrote a 2-D magnetostatic FEM from the weak form** (scikit-fem, gmsh, 141k elements) to check it independently: agreement to **0.07 %**.
- Closed the system at **77 kg dry**, and **published every defect found** — 46 numbered open problems, with acceptance bands declared before each analysis ran. **Retracted my own claims when evidence went against them**: an independent propagator (GMAT) falsified a claim in the paper's abstract; my cost model contradicted the paper on which parts dominate; a literature review found prior art I had missed. Presented at **DRDO ARDE**, Pune (2021) and the **India Science Festival**, IISER Pune (2024).

Electromagnetic Launch System for Vertical Silo-Based Deployment

Sole author • feasibility study, IEEE format 2025

- Applied the same magazine-fed electromagnetic launch architecture to a 500 kg booster class: belt-feed indexing, pulsed coilgun ejection, ignition deferred to 20 m altitude. **2.50–4.17 MJ per launch** across a sourced 15–25 % efficiency band, **10.6–15.0 cycles/min** on a 1 MW supply.
- **Selected a coilgun here and rejected one for the CubeSat deployer** — the same trade, resolved oppositely, because both are sized by the payload's g-limit via $s_{\min} = v^2/2ng$. A 15 m rail holds 8.5 g against a 10–15 g propellant grain; a 1.5 m track holds 10.7 g against a 14 g CubeSat. Two machines, one constraint.

Re-entry Parachute Deployment System

IREC 2025, Team THRUST • Midland, Texas 2025

- **2nd place globally**, SDL Payload Challenge. Modelled aerodynamic drag, engineered structural reliability into the recovery mechanism, and validated the deployment sequence against failure modes (FMEA).

Powertrain and mechanical

Powertrain Engineer

Wrench Wielders Racing — SIT Formula Student 2025–present

- Engine teardown, inspection and reassembly on the KTM 390 single-cylinder platform; ECU calibration, fuel mapping and throttle response; passed SAE India technical inspection. Drivetrain integration and packaging — sprocket sizing, engine-to-chassis mounting, and weekly coordination with chassis, suspension and aero on weight distribution.

Founder & Powertrain Lead

Poona Motor Club, Pune 2021–present

- Raised output **46 → ∼60 bhp** on an RE Super Meteor 650, **dyno-validated across the band**, by re-engineering combustion airflow (head porting, camshaft profiling, high-flow induction, exhaust scavenging) and rewriting Powertronic fuel and ignition maps for target AFR at every throttle position. Replicated on a client Interceptor 650.
- **Reverse-engineered Powertronic's obfuscated map binary** and built a dual-map editor so two calibrations could be compared side by side on the bike.
- Standardised diagnosis and tuning across four platforms (RE Sherpa 450, KTM 390, CB600RR, Duke 390); ground-up rebuild of a 2009 TVS Apache 180, ANSYS-validated chassis.

Two-Stage Gear Train Supercharger — **completed** (previous final-year project, closed out in third year). Designed and metal-fabricated a two-stage spur gear train giving **1:5 RPM multiplication** within cleared gear-stress margins, letting a standard high-torque DC motor replace an ultra-high-speed unit; FMEA completed (ANSYS, MATLAB).

Applied AI and software

AI System Architect

Avisys Services • AI Associate, Sep 2025 – Jan 2026 Jan 2026–present

- Own AI systems architecture and product design for AviateCX, an enterprise telecom CRM: shipped a **RAG** retrieval assistant (pipeline, vector store, prompt engineering) and a role-based dashboard across all four user tiers. Also architecting a multi-state Resource Adequacy compliance platform for Indian power utilities (GridPath, pandapower, Antares under a versioned regulatory-rules engine).

Skills

Simulation: scikit-fem, gmsh, GetDP, CalculiX, ngspice, GMAT, magpylib — magnetostatic FEM, structural FEA, circuit, orbit propagation. **CAD/CAE:** Fusion 360, SolidWorks, AutoCAD, ANSYS. **Programming:** Python (numpy/scipy), MATLAB, C, LaTeX, git. **Hardware:** fabrication, 3-D printing, CNC, ECU calibration, dyno testing, FMEA. Coursework in Finite Element Methods and CFD. Every skill here is mapped to the file that evidences it: docs/SKILLS.md.